Study Room Sniper

P2 analysis and high-level design

Availability alerts for UT Dallas students seeking study rooms at Eugene McDermott Library.

TEAM 4 · SECTION 003 · v1.0 CANDIDATE

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Reading guide

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Revision and P1 feedback

Team 4, Section 003 · CE/CS/SE 3354 · 3 November 2026

Revision	Record	Changes
v0.1 · 18 October	P1 proposal	Defined the room watch, alert conditions, 23 requirements and four use cases.
v1.0 candidate · 3 November	P2 pull request #19	Traces the same requirements through the design. Adds a polling experiment, a student survey and a plan for testing the service.

Response to P1 feedback

Rubric item	Changes in P2	Next step
C1 · Need and feasibility · 6.5/8	The survey asks what students did the last time they couldn't find a room. The spike measures how often one reader can check 17 rooms and how long an alert takes.	Pilot the survey with at least ten enrolled students. Ask the library which source and read rate it supports.
C2 · Requirements · 10/10	The acceptance checks retain the stub tests and numerical thresholds. Section 8 gives results for the request cap and simulated alert delay.	Measure the full delay at a test mailbox. Run the timed watch-creation task and accessibility walkthroughs.
C3 · Use cases and traceability · 8/8	UC-01 and UC-02 now have scenario tables and sequence diagrams. UC-03 follows a student through cancellation. Section 9 links all 23 IDs to the design.	Have a teammate trace one cancellation and one alert from the student action to the recorded outcome.
C4 · Team plan and repository · 3.5/6	The report, editable sections and replay are together in PR #19. Section 10 proposes a Wednesday meeting and GitHub Issues for team decisions.	Each member commits their section or findings and asks a teammate to review the change.
C5 · Individual contribution · 4/8	The student survey and its collection instructions are in the survey commit.	Record the pilot dates, response counts and any change students suggest to the watch settings.

The P1 team score was **28/32**.

Contribution evidence

P2 change evidence

Halogen419 opened PR #19 on 3 November with the five commits below.

Commit	Issue	Files and decisions
b4a7b98	#14	Availability replay, output and reproduction instructions.
d0f955c	#15	Use cases, five diagrams, four architecture decisions and requirement traceability.
62dd643	#12	Requirements, constraints and their acceptance checks.
28cad85	#4	Survey questions, collection instructions, working agreement and P3 schedule.
0e1a232	#4	P2 report, contribution record and README.

Team responsibilities

The P1 roles carry through to the review. The proposed tasks below identify who handles each follow-up. Completed changes are recorded in the PR commit history.

Member	Area	Next contribution
Alexander Eng	Change control	Record the review decisions and the commit the team accepts as v1.0.
Arshan Khimani	Feasibility	Compare the library's supported response with the fields and request budget used in the replay.
Lee Baillard	Elicitation	Summarize the library and OIT replies with dates and affected requirement IDs.
Mayand Dhami	Quality and accessibility	Pilot the survey, revise any unclear questions and publish the aggregate counts.
Henry McGahan	Use cases and traceability	Walk through a watch cancelled between matching and dispatch. Check the sequence and trace against FR-09.
Isaam Memon	Risk and integration	Rerun the spike from a clean checkout and review the assembled report.

The room search

Study Room Sniper helps an enrolled UT Dallas student find a study room at Eugene McDermott Library. Today a student who finds the rooms occupied must revisit the library calendar to see whether a cancellation or a lapsed hold has freed a slot. The student saves a date, time window and group size once. When a matching slot appears, the service sends a link to the library calendar. The student makes any reservation there.

The P1 baseline defines the watch and its acceptance criteria. The constraints cover NetID, read-only library access and cost. The verification guide gives the checks for each requirement.

Scope and vocabulary

The responsive web service covers the library's nine group study rooms and eight smart rooms. Students use NetID, manage their own watches, receive email, and can opt into browser push. Operators can inspect read health. The P1 scope excludes booking or altering reservations, the One Button Studio, other locations, SMS, native apps, predictions and room recommendations.

Term	Meaning in this baseline
Watch	One student's saved date, time window and minimum group size. Its state is Active, Cancelled or Expired. An alert leaves the watch Active.
Slot and room	A slot is a library-published time interval for one room. The room's listed capacity determines whether the group will fit.
Bookable	Provisionally, the library shows the slot as free. Q4 asks whether a student's own reservation limits must also be considered. The library determines final eligibility when the student opens the link.
Match	A free slot on the requested date, wholly inside an Active watch's window, in a room with enough capacity.
Read cycle	One read-only sweep of all in-scope rooms. The request cap applies across the service, including reads that confirm a match.
Alert	Email describing matches for a student, with optional browser push. Delivery time and whether the student notices the alert are separate observations.
Operator	Team member who monitors the source reader and investigates failures.

P1 draws on ISO/IEC/IEEE 29148:2018 for requirements-writing style (R1), the library inventory and policies (R2–R4), W3C WCAG 2.2 (R5), and the course syllabus and P1 template on Canvas (R6). The library statements below carry the P1 interpretation pending Q1–Q4. R2, library calendar · R3, reservation policy · R4, room policies · R5, WCAG 2.2

Stakeholders and feasibility

The team controls the watch interface, watch state, periodic reader, matching, alert dispatch and operator view. The library owns room inventory, policy, availability and booking. OIT owns the NetID sign-in service. Email and web push providers deliver messages. The application never requests a student's library password.

Stakeholder	Need and evidence status
Students	Set up a watch quickly on a phone, receive enough detail to act and stop alerts when plans change. The pilot asks students how often they check again and what email delay they could use (Q5).
Students using assistive technology	Complete watch creation and watch management with keyboard or screen reader and receive information independent of color or sound. P1 references WCAG. Direct consultation and campus guidance remain open under Q6.
Library reservations staff	Keep library traffic within an acceptable rate and preserve fair access. P1 inferred this from published policies. The library's approval of the source and rate remains open.
UTD OIT	Approve NetID integration and the attributes a student project can receive. Q3 and Q6 require an OIT or campus accessibility response.
Team and operators	Work within a zero-cost budget and detect stale reads and failures. The operator view shows when a read last succeeded.

Assumptions and feasibility at P2

P1 ID	Working assumption	Evidence and consequence
A1	A program can read timestamped slots without student credentials.	The public list identifies rooms. The lookup did not establish a supported slot feed. Until Q1 is answered, FR-03 and FR-04 can be demonstrated against fixtures.
A2	The library permits automated reads at up to 20 requests/minute total.	The limit is a ceiling set by P1. Ask for the library's approved rate under Q2 and pause if requested. A lower rate may change achievable alert delay.
A3	OIT approves NetID sign-in and supplies NetID and email.	Obtain approved attributes under Q3. If access is refused, changing CON-01 needs a change request. Email verification would lose the original enrollment assurance.
A4	The 17-room inventory remains small, with at most 20 rooms anticipated.	The spike exercised 17 rooms. Future inventory and how many dates one response covers determine whether FR-03 fits QR-05.
A5	Students notice an alert within a few minutes on a phone.	A simulated or real delivery timestamp cannot measure attention. Ask students under Q5 before revising channels or thresholds.
A6	The library can supply a reliable listed capacity per room.	Confirm the source and reconcile all 17 room IDs. Dropping group-size matching would require a change to FR-01 and FR-04.

The offline spike supports a shared reader under its fixture assumptions. Live operation depends on library access, OIT approval and measured provider delivery. No-cost plans have quotas, and 100 simulated recipients in one day could consume one candidate email plan's full daily allowance. These gates affect the P3 schedule. Spike method and results

Functional requirements

The IDs, priorities and thresholds below carry forward P1. The use-case key is UC-01 create a room watch, UC-02 receive an alert, UC-03 manage watches, and UC-04 monitor service health. The acceptance column states what the reviewer should observe. Step-by-step checks · Design traceability

ID and priority	Student or operator outcome	Acceptance and use case
FR-01 · MUST	As a signed-in student, I can save a date within the next seven days, a time window and minimum group size so I can stop refreshing the calendar.	The saved watch displays all entered criteria. A date more than seven days ahead is refused with a message. UC-01.
FR-02 · MUST	As a student, I sign in with UTD NetID before I can create, see or cancel my watches.	Each signed-out action requires NetID sign-in. The action becomes available after successful sign-in. UC-01, UC-03.
FR-03 · MUST	While the library is open, the service reads availability for every in-scope room at least once every two minutes.	Against a reachable stub, the request log for a 60-minute run shows no room with a read gap above 120 seconds. UC-02.
FR-04 · MUST	As a student, I receive matches only for free slots on my date, fully inside my window and in a room that fits my group.	Release one fitting slot and one outside the window. Within one read cycle only the fitting slot is matched. UC-01, UC-02.
FR-05 · SHOULD	Before sending an alert, the service confirms that the matching slot is still bookable.	Withdraw a slot between discovery and confirmation. Its owner receives no alert for it. UC-02.
FR-06 · MUST	As a student, I receive email covering every match found in a read cycle.	Release a matching slot. The owning student's test inbox receives exactly one alert. UC-02.
FR-07 · MUST	I can identify the room and time from the alert and open the library booking page.	Received alert contains room name, slot start and end, and a working library booking link. UC-02.
FR-08 · SHOULD	I can choose browser push as an additional channel for a watch.	Release a match for each of two watches. The opted-in watch receives email and push, the other email only. UC-01, UC-02.
FR-09 · MUST	I can cancel an Active watch.	After cancellation the watch shows Cancelled, and a later matching release produces no alert. UC-03.
FR-10 · MUST	I can list my own watches and see Active, Expired or Cancelled status.	In a two-account check, each owner sees only their watches with the correct states. UC-01, UC-03.
FR-11 · MUST	When the watch window ends, the service marks it Expired and stops evaluating it.	Advance the clock past the end. Within five minutes it shows Expired. A later matching slot produces no alert. UC-02, UC-03.
FR-12 · MUST	A watch receives at most one alert for a given slot.	Keep one slot free across ten cycles. Its owner receives exactly one alert for that watch and slot. UC-02.
FR-13 · SHOULD	A student can hold at most five Active watches.	With five Active watches, attempting a sixth is refused with a message. UC-01.
FR-14 · COULD	As an operator, I can see the last successful availability read and the number of failed reads in the last hour.	Stop the stub for ten minutes. The timestamp freezes and failures rise above zero. Reads resume after restoration and failures age out of the rolling hour. UC-04.

Quality requirements

ID and priority	Requirement	Acceptance method
QR-01 · MUST	At up to 200 Active watches, at least 95% of alerts reach the student's email service within 180 seconds of a slot becoming bookable.	Compare release timestamps with delivery logs over 100 scripted releases. At least 95 must meet the deadline.
QR-02 · SHOULD	The service is available at least 99% of library opening hours over a continuous 14-day trial.	Probe every five minutes and divide successful opening-hours checks by all scheduled opening-hours checks.
QR-03 · MUST	At least 90% of first-time users create a watch on a phone in under 90 seconds without help.	Moderate at least ten first-time UTD students, timing first screen to watch appearing in the list. At ten participants, at least nine must succeed.
QR-04 · MUST	Every screen used in UC-01 and UC-03 meets WCAG 2.2 Level AA with zero serious or critical violations.	Scan every screen and complete every step in both cases using keyboard-only and screen-reader walkthroughs.
QR-05 · MUST	Total traffic to the library reservation system stays at or below 20 requests per minute, independent of the number of Active watches.	Inspect the 24-hour request log with 200 Active watches, including confirmation and retry traffic. No minute may exceed 20.
QR-06 · MUST	The only personal data stored about a student are NetID, email address and watches.	Review full exports for three test accounts against those permitted categories.

3.3 Constraints and baseline relationships

CON-01 governs student sign-in, CON-02 protects the library's reservation boundary, and CON-03 binds the project to free services. Their sources, effects and verification are documented separately. Constraints

FR-14 is the sole functional requirement for UC-04. FR-11 covers expiry discovered during alert processing as well as the list state seen by a student. FR-13 reaches the refusal flow in UC-01. The P2 matrix extends the original functional links to cover architecture, quality requirements and constraints. Traceability matrix

Constraints on the design

These three P1 constraints bind every proposed design. The team must record and review any relaxation through its change-control process. P1 requirements · Verification

CON-01 · UTD NetID sign-in only

UTD NetID single sign-on is the only student sign-in mechanism. P1 attributes this to an assumed campus requirement. Q3 asks OIT whether the project may register and which attributes establish identity and provide an email address. Only the student's own verified session may create, list or cancel that student's watches.

P3 can exercise the identity boundary with synthetic principals while OIT advice is pending. A live pilot requires approved settings and claims. Replacing NetID with an email-verified account would require an approved baseline change. Verify the three protected operations signed out and as two different signed-in students, and inspect the registered identity-provider configuration. Issue #9

CON-02 · Read-only library access

The service may read permitted availability. It must never create, change or cancel a reservation or hold a student's library credentials. The constraint comes from the P1 team decision and the library reservation policy. Q1/Q2 concern the source and permission to automate reads. Students finish any booking on the library's own site.

The reader accepts only approved availability operations. A booking link transfers control to the student's browser. If the source changes or is withdrawn, pause live reads and use fixtures while the team reviews the impact. Verify the outbound operation log through matching, confirmation and cancellation, including an attempted unsupported operation against a stub. Issue #10

CON-03 · Zero cost to the team

All services used to build and run the project must cost the team nothing. This is a team budget decision. The design must account for hosting, storage, email and optional push, including retries and expected peak volume. A free tier can be exhausted even during a small pilot.

Record provider limits, account conditions and expected usage before enabling delivery. Under a hard quota, surface exhaustion and pause affected sends instead of allowing a paid overage. Waiting for quota renewal can violate an alert's delivery deadline and must be recorded under QR-01. Verify the configuration, expected volume, quota behavior and actual cost for the proposed pilot. Issue #11 · Quota observations

UC-01 · Create a room watch

Students create watches, receive alerts and manage their own watches in UC-01 through UC-03. They book on McDermott Library's site. In UC-04, the operator can see when a library read failed and when the last one succeeded (FR-14).

Primary actor Enrolled student. **Supporting actor** UTD NetID SSO. **Entry condition** The service is available and the student has a NetID. **Trigger** No suitable room is available for the student's planned session.

Turn	Action and observable response
0	The student opens Study Room Sniper to request an alert.
1	The system requests NetID sign-in when a verified session is absent. SSO returns the student's identity and the system opens the student's own watch list.
2	The student selects **New watch**. The system presents fields for a date, start and end times, minimum group size and optional browser push. Email is selected by default.
3	The student submits the criteria. The system checks the date against the seven-day horizon, the order of the times and a positive whole group size.
4	The system checks the owner's Active count and compares the criteria with the latest successful availability read. In the main flow, there is room for another watch and no current match.
5	The system saves one Active watch for this owner and acknowledges the submission by showing its criteria, channels and **Active** status in the list. The student can now leave the page.

Alternate turns. At 1a, cancelled or failed sign-in returns the student to sign-in without creating a watch. At 3a, an invalid date, empty or reversed window, or invalid group size leaves the entered values in the form with a field-specific correction. At 3b, denied push permission keeps email selected and explains the fallback. At 4a, a sixth Active watch is refused, including when two submissions compete for the last place. The student can cancel a watch in UC-03 and try again. At 4b, a slot already shown free is presented with the library link. The student chooses whether to save the watch. Declining ends the flow with no new watch. If there is no usable read yet, acknowledge the saved watch and wait for the next successful cycle.

Exit and trace. A successful save belongs to the signed-in student and is eligible for the next read cycle. FR-01, FR-02, FR-04, FR-08, FR-10, FR-13.

UC-02 · Receive an alert

Primary actor Student. **Supporting actors** Library Reservation System (LRS), email service and optional browser push service. **Entry condition** At least one Active watch has time remaining and the library is open. **Trigger** A fitting slot becomes free. Until Q4 is resolved, “bookable” means the library shows the slot as free. Only the library can establish whether this student may reserve it.

Turn	Action and observable response
0	A slot opens in the LRS. A student with an Active watch is waiting for an alert.
1	On the next scheduled cycle the system reads all in-scope rooms. The LRS returns room availability. A successful cycle records when the read completed.
2	The system selects free slots on the watch's date, fully inside its time window, in rooms with listed capacity at least the group size. It excludes watch/slot pairs already alerted or pending.
3	The system requests a fresh, read-only confirmation under the shared budget. The LRS still shows the slot as free. The system checks the watch is still Active and in time and claims the eligible pair once.
4	The system groups this cycle's confirmed matches for the student. The email gives each room, start, end and library booking link. Optional push covers only matches from push-enabled watches. Each channel reports its own result.
5	The system records channel acceptance. The student receives the alert and follows the link. The library shows the current booking page and handles any reservation the student chooses to make.

Alternate turns. At 1a, an unreadable or failed cycle is recorded for UC-04. The system discards its incomplete data and tries the next scheduled cycle. At 2a, a watch whose end has passed becomes Expired and is skipped. At 2b, simultaneous matches are listed together. A repeated slot is listed once even if it satisfies two of the same student's watches, with separate watch/slot records to prevent re-alerts. At 3a, a lost slot, unavailable confirmation or newly cancelled watch yields no send for that pair. A later read can reconsider an eligible, unsent pair. If dispatch is delayed or retried, confirm again before handoff. At 4a, explicit channel rejection allows up to three retries within five minutes. An accepted channel is skipped on retry. An unknown outcome is recorded for reconciliation. At 5a, a slot taken before the student opens the link is unavailable at the library. The watch remains Active for other openings.

Exit and trace. On success, the required email is accepted and its slots are recorded against the corresponding watches. A watch remains Active until cancellation or expiry. FR-03 through FR-08, FR-11, FR-12.

UC-03 · Review and cancel

Primary actor Student. **Supporting actor** UTD NetID SSO. **Entry condition** The student has a NetID. Cancellation requires an Active watch owned by the student. **Trigger** The student wants to check a watch or stop future alerts for it.

Turn	Action and observable response
0	The student opens **My watches**.
1	The system resolves the student's NetID, requesting sign-in if needed. It returns only that owner's watches and shows each status in text.
2	The student chooses **Cancel** on an Active watch. The system identifies the date and time window of the selected watch and asks for confirmation.
3	The student confirms. The system checks ownership and current status again and records the Active-to-Cancelled transition.
4	The system acknowledges cancellation and displays the row as **Cancelled**. Later matches for this watch produce no new dispatch.

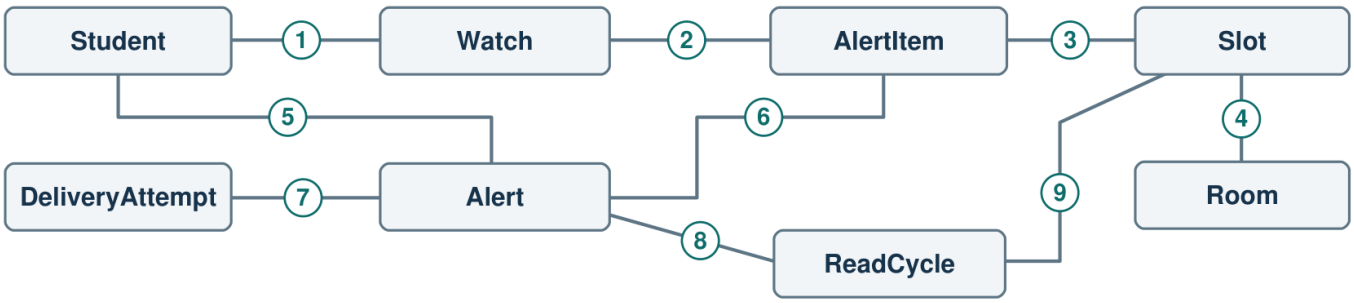
Alternate turns. At 1a, failed sign-in reveals no watch data. An empty list provides a link to UC-01. At 2a, the student can finish after reading the list or abandon the confirmation. At 3a, a second cancellation returns the current Cancelled state. A watch whose window has already ended shows Expired. A request referring to another owner's watch receives no details and changes nothing. A provider may deliver an alert accepted before cancellation. Turn 4 promises only that subsequent dispatches stop.

Exit and trace. Viewing changes no active watch criteria. A cancelled watch no longer uses one of the student's five Active places. FR-02, FR-09, FR-10, FR-11.

Self-review log

Review of this draft	Finding and resulting change
UC-01 against P1 §3.4	The initial success path alone missed the already-free case. Added 4b and made the student's save decision explicit.
UC-02 against FR-06, FR-08 and FR-12	Separate optional push from required email. Record each channel's acceptance and retain the watch/slot association when batching.
UC-03 against FR-09 and FR-10	Scope the list and cancellation to the signed-in owner. Added the stale-tab, repeat-cancel and already-accepted-alert cases.

Conceptual domain model



1. 1 Student owns 0..* Watches
2. 1 Watch motivates 0..* AlertItems
3. 1 Slot appears in 0..* AlertItems
4. 1 Room offers 0..* Slots
5. 1 Student receives 0..* Alerts
6. 1 Alert includes 1..* AlertItems
7. 1 Alert tracks 0..* DeliveryAttempts
8. 1 ReadCycle originates 0..* Alerts
9. 0..* ReadCycles observe 0..* Slots

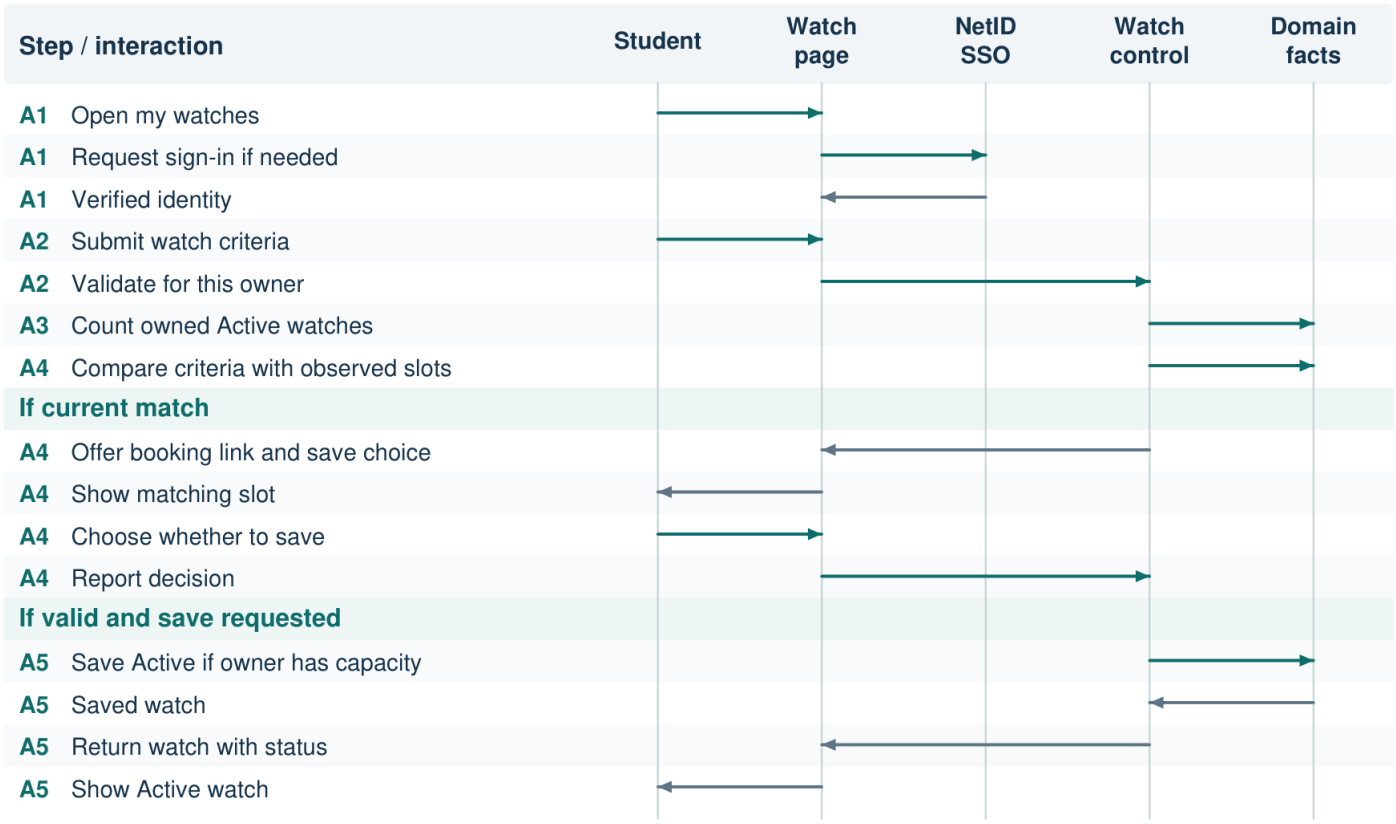
The **Student** is identified by NetID and has an email address. A **Watch** carries a local date, window, minimum group size, channel choice and status. A **Room** has a library identity and listed capacity. A **Slot** denotes one room and one start/end pair. Availability is the state of a slot observed during a **ReadCycle**. A failed cycle can have zero usable observations. **AlertItem** records a qualifying watch/slot pair, unique across cycles for that watch. An **Alert** groups items for one owner and one originating cycle. A **DeliveryAttempt** records channel and outcome, so one channel can fail after another succeeds. These are domain concepts. Storage choices belong in §6.

Worked match. A watch for four students on 6 November, 14:00–16:00, accepts a free four-seat room from 15:00–16:00 that day. A 15:30–16:30 slot extends past the end. A three-seat room fails even if its time fits. If the accepted slot appears in ten reads, the existing watch/slot item suppresses a second alert for that watch.

UC-01 · Scenario and sequence

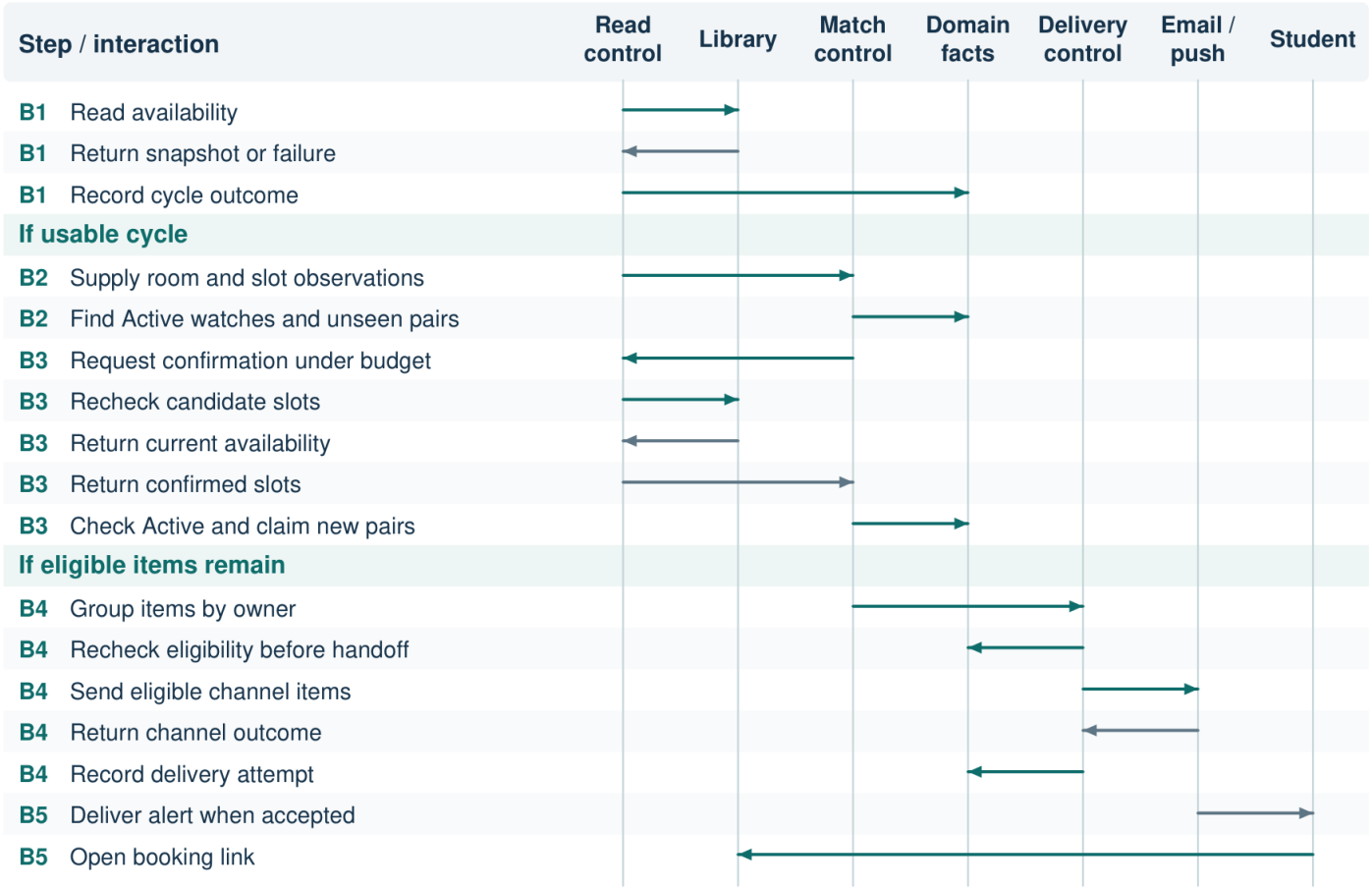
Each row describes a responsibility. Boundaries exchange information, controls coordinate the goal and entities carry domain facts. The sequence uses the same row keys as message labels.

Key and turn	Actor or event	Boundary	Control responsibility	Entity or outcome
A1 · 1	Student opens watches	Watch page / NetID SSO	Establish the verified owner	Student identity
A2 · 3	Student submits criteria	Watch form	Validate date, window, group size and channels	Proposed watch
A3 · 4, 4a	Valid submission	Watch form	Check owner capacity and reject a sixth Active watch	Student and owned Watches
A4 · 4, 4b	Criteria ready	Watch form	Compare latest usable slots. If one fits, offer its link and a save choice	Room, Slot, ReadCycle
A5 · 5	Submission may proceed	Watch list	Recheck limit when saving and acknowledge	New owned Watch, Active



UC-02 · Scenario and sequence

Key and turn	Actor or event	Boundary	Control responsibility	Entity or outcome
B1 · 1, 1a	Poll due	LRS reader	Read in-scope rooms under budget and record outcome	ReadCycle, Room, Slot
B2 · 2, 2a–b	Successful cycle	Availability view	Compare each current watch, suppress repeats and group candidates by owner	Watch, Slot, AlertItem
B3 · 3, 3a	Candidate remains	LRS reader	Confirm current library state, recheck the watch and claim each eligible pair once	Watch, Slot, AlertItem
B4 · 4, 4a	Confirmed items	Email / push services	Hand off the matching slots per eligible channel and record outcome	Alert, DeliveryAttempt
B5 · 5, 5a	Student opens link	Library booking page	Library decides current eligibility and availability	No reservation in our domain



Six consistency checks

Check	Cross-check and result
1. Actors and boundary	A1 uses the external NetID actor in UC-01 and UC-03. B1/B3 only read the LRS. Only the student follows the link at B5, consistent with CON-01 and CON-02.
2. Step coverage	UC-01's sign-in, validation, capacity, current-match choice and save appear at A1–A5. UC-02's read, match, confirmation, handoff and acknowledgment appear at B1–B5.
3. Vocabulary and associations	A3 counts only Watches owned by one Student. B2 compares each Watch with a Slot in its Room. B4 follows AlertItem back to the motivating Watch and the exact Slot shown.
4. Multiplicity and uniqueness	A5 enforces at most five Active watches, while historical Watch multiplicity remains unbounded. B3 claims a given watch/slot once. One Alert can cover several distinct items.
5. State and time	Creation enters Active. UC-03 can change Active to Cancelled. An ended Active watch moves to Expired. B2/B3/B4 exclude ended or cancelled watches. A successful alert leaves the watch Active.
6. Failure and visibility	B1 failure creates a ReadCycle outcome usable by UC-04 and stops evaluation of that cycle. B4 explicit rejection creates a DeliveryAttempt and permits bounded retry for that channel. Neither path shows a successful delivery solely because a match was found.

Quality-attribute scenarios

These are acceptance targets from P1. Section 8 identifies which of them the spike exercises and under what assumptions.

Scenario	Source, stimulus and operating conditions	Affected part and response	Response measure
QA-1 Timely alert (QR-01)	The stub releases a matching slot while 200 watches are Active. The library is open and the slot remains free.	C3 records availability, C4 identifies the watch and C5 sends the confirmed match. Measure from release until the student's email service accepts the message.	At least 95 of 100 scripted releases reach the email service within 180 s. Opening the message is outside this clock.
QA-2 Bounded traffic (FR-03, QR-05)	Many watches match at once during a 24-hour run with 200 Active watches. Confirmations compete with scheduled reads.	C3 arbitrates all library requests. It shares each room read across watches, coalesces confirmations of the same room where the source supports it, and defers work when the budget is exhausted.	At most 20 outgoing requests in any rolling 60 s. Separately, a 60-minute run against an always-reachable stub must show a maximum read gap of 120 s for every in-scope room.
QA-3 Service availability (QR-02)	An automated client probes the service every five minutes during library opening hours over 14 consecutive days.	The web process serves the check. C6 separately reports library-read health so an old snapshot remains visible to the operator as stale.	Successful checks divided by all scheduled opening-hour checks must be at least 99%. Count missing checks as failures.
QA-4 Accessible watch creation (QR-03, QR-04)	A first-time student opens the watch form on a phone. A keyboard or screen-reader user creates and cancels a watch.	C1 supplies labeled controls, announces validation and status changes in text, and returns focus to the affected control or result.	At least 9 of 10 first-time UTD students finish unaided in under 90 s. UC-01 and UC-03 screens must meet WCAG 2.2 AA, have zero serious or critical scan findings, and support every walkthrough step.

Application components

A modular monolith serves the watch list and runs one background reader. The same release contains both, with one transactional store. A student who opens the watch list reads stored watch state. Their visit generates no extra request to the library. ADR-01 records why a service per component and polling from each student's browser were rejected.

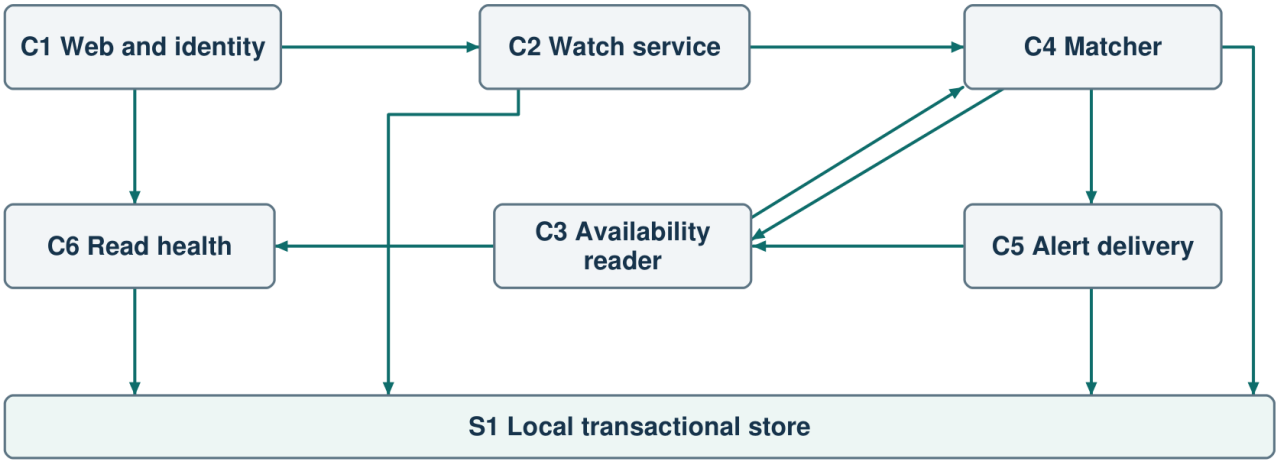


Figure 6.1. Arrows show which component calls or supplies work to another. C3 supplies a shared snapshot to C4. C4 requests a fresh confirmation from C3 before claiming a pair, and C5 requests a new confirmation on a delayed send or retry. The C2 to C4 path checks the latest usable snapshot for the already-free alternative in UC-01. S1 is private to the application.

Component	Analysis concepts and responsibility
C1 Web and identity	Presents Student and Watch data, verifies the campus sign-in response and passes the authenticated subject to C2. The operator view requires a separate role.
C2 Watch service	Owns Student and Watch. Filters reads and cancellations by authenticated owner. A single transaction enforces the five-Active-watch check and creation. Expiry continues even while the library is closed.
C3 Availability reader	Owns the read side of Room, Slot and ReadCycle. One scheduler and one rolling request budget cover polls, retries and confirmations. Rejects incomplete or malformed source data.
C4 Matcher	Compares Active Watch criteria with Slot times, status and Room capacity. Confirms candidates through C3, checks eligibility and claims each new AlertItem by watch and canonical slot.
C5 Alert delivery	Owns Alert and DeliveryAttempt. Checks eligibility at handoff, groups surviving items into one email per student per cycle, sends push for eligible items only, and records each channel's outcome.
C6 Read health	Consumes ReadCycle results. Reports the last complete successful read and a rolling one-hour count of failed reads to an authorized operator. A failed cycle leaves the prior success time intact.
S1 Local store	Persists ownership, watch state, alert items, delivery outcomes and read-health timestamps. A uniqueness constraint on watch and slot protects the alert history across worker restarts.

Deployment and trust boundaries

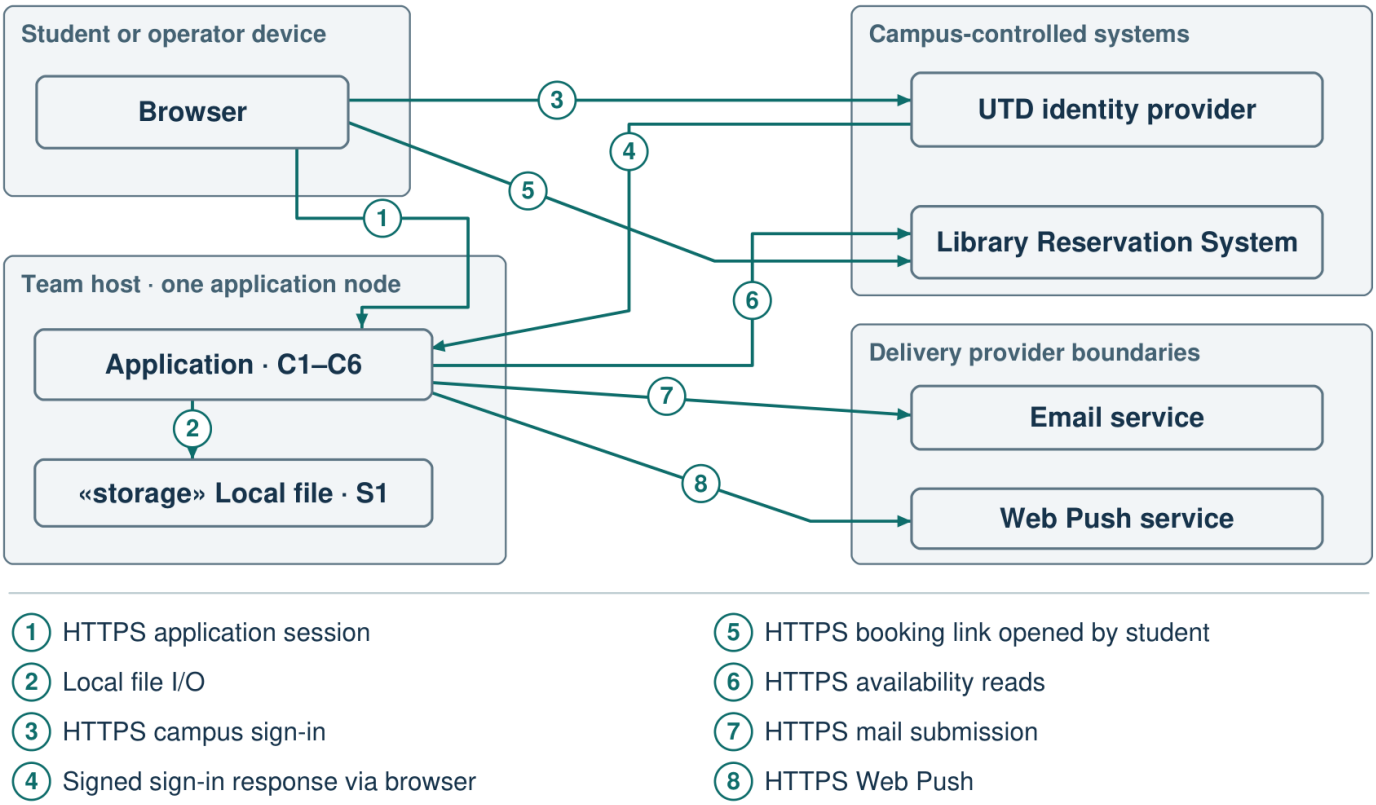


Figure 6.2. The browser is an untrusted client. The application derives watch ownership from the verified campus subject. Only the browser follows the booking link. The read adapter targets an approved availability resource and exposes no reservation operations. The mail provider receives the destination and the match details needed for delivery. Push requires the student's opt-in and a privacy decision under QR-06. The sign-in exchange is conceptual until OIT confirms the supported protocol and claims under Q3.

Design principles in use. Ownership checks sit in C2 so the list and cancellation screens use the same rule. Isolating the LRS format in C3 lets a library format change fail a read without expiring a watch or sending a false alert. C4 compares domain data without calling a provider, so the same match can be repeated against a fixture. C5 keeps per-channel outcomes so a successful email does not get replayed when push fails.

Choosing a reader architecture

Three limits shape the design: the library request budget, the time a student can wait for an alert, and the team's zero-cost budget.

ADR-01 Keep one deployable and one background reader

Choice. Run the web interface and scheduled work as modules of one application using a private transactional store. The common store makes watch cancellation, ownership and alert claims visible to the worker. A single active scheduler owns the library request budget.

Alternatives and cost. Independently deployed web, polling and delivery services would allow separate scaling. They would also need a queue, shared rate coordination and recoverable messages between services. Browser polling would reduce server work, but each open tab would compete for the 20-request budget, and closing the browser would stop email watches. The proposed host has a single failure point and a restart can delay a cycle. Revisit if the 14-day QR-02 trial misses 99% or the platform cannot keep a worker awake at zero cost.

ADR-02 Share room reads and budget confirmations

Choice. Poll rooms on a common schedule and match the results locally against all Active watches. Put every library request through the same rolling-minute limiter, including a fresh confirmation before each alert batch. Share confirmations across candidate watches for the same room when one read covers those slots. An unreadable confirmation suppresses the affected item. A later cycle may discover it again.

Alternatives and cost. Reading separately for each watch makes library traffic grow with the number of students. Reusing the old snapshot as final confirmation saves requests but misses a slot taken between discovery and sending. With one response covering the required horizon of one room, 17 rooms require at least 17 scheduled reads per 120 s. Confirming each room once more in that interval requires another 17 of the 40 available requests. Timing within each minute still matters. Under pressure, protect scheduled reads, defer excess confirmations, and count delayed alerts against QR-01. If the source requires one call per room per date, seven dates would require 119 scheduled reads per 120 s before any confirmation. Reopen the decision if Q1 identifies that contract or Q2 sets a lower cap.

Delivery and identity decisions

ADR-03 Keep a durable identity for each alert item

Choice. C4 confirms candidate slots through C3 and claims each eligible watch/slot pair once in S1 using the library room identifier and normalized slot start and end. A logical alert groups claimed items by recipient and cycle. Its email lists a room and time once even when two of that student's watches qualify. Push lists only items from opted-in watches. C5 records outcomes separately by channel and rechecks ownership, Active status and window end at dispatch. A delayed send or retry requires a fresh C3 confirmation. Cancelling a watch suppresses unsent items for that watch. A message already handed off to a provider cannot be recalled.

Alternatives and cost. An in-memory “already sent” set loses history on restart. Sending before recording leaves a crash window that can duplicate an email. Keep the item and attempt outcome durably, including a pending attempt before handing off. Retry a definite provider rejection at most three times within five minutes, with fresh checks. If a timeout or crash leaves acceptance unknown, reconcile using a provider receipt or idempotency key where supported. Otherwise record the unresolved attempt and suppress an automatic resend. This preserves FR-12 at the cost of a possible missed FR-06 delivery. Release an unsent item claim after a slot fails confirmation so a later genuine reopening remains eligible. Revisit if the provider cannot distinguish rejection from unknown acceptance often enough to meet QR-01.

ADR-04 Bind every watch operation to a campus identity

Choice. Validate the UTD sign-in response in C1 and derive the owner in C2 from its verified subject. Store NetID, email and watch data. Keep operating logs free of student names, addresses, IPs, raw sign-in claims and full notification payloads. Associate deduplication records with the watch that generated them. Use role checks for the read-health view and keep health counters free of student data.

Alternatives and cost. Local passwords would require extra credentials and a change to CON-01. Trusting an owner ID supplied by the browser would let one student read or cancel another student's watch. NetID-only access depends on OIT approval and the claims OIT releases. Browser push ordinarily needs a stored endpoint and keys linked to the user, which QR-06 does not explicitly permit. Keep this delivery path gated pending a documented privacy decision or approved change. Email remains the required channel.

7.1 Trade-offs to revisit

Pressure	What the choice protects	Cost the team must measure or resolve
Faster discovery versus library load	One global budget caps scheduled reads, retries and rechecks.	A tighter library cap, paging or a long confirmation queue may push email past 180 s.
Delivery coverage versus duplicate messages	Durable item claims and per-channel receipts protect students from repeated alerts.	An ambiguous provider response can leave a student without an email until reconciled.
Independent scale versus zero cost	One host avoids a distributed queue and keeps the budget in one place.	Host sleep or downtime affects both the web interface and scheduled reads.
More delivery options versus minimal student data	Email works with the P1 data allowance.	Push needs an explicit decision on endpoints, keys and deletion before enabling it.

Can the reader find a slot in time?

The reader has to revisit every room within two minutes and make at most 20 library requests per rolling minute, including confirmations. This replay tests whether one shared reader can do both with 200 Active watches.

What the public pages showed

On 3 November the public library index listed nine group rooms and eight smart rooms. The group-room page returned HTTP 429, the all-rooms page could not be retrieved, and the smart-room page showed the calendar legend without dated slots in the retrieved text. The replay therefore uses simulated openings. The library still needs to confirm how to read live slots and at what rate. Library index · Group rooms · Smart rooms · All rooms

Reproduce the experiment

From this directory, run the command below with Python 3.10 or later. It uses only the standard library. The JSON output should match the recorded run. A failed check raises an error.

```
python replay.py
```

Over one simulated day, 100 slots open in five groups of 20. Each slot fits one watch. A second watch requests the same time for a group too large for the room. One extra slot disappears between the first read and confirmation. The reader polls a room every six seconds and checks it again three seconds later if it finds new matches.

Measurement	Observed in the replay
Scheduled room polls / extra confirmation reads	14,400 / 86
Longest gap between consecutive polls of the same room	102 s
Peak requests in any rolling 60 s, confirmations included	20
Eligible persistent watch/slot pairs alerted / fleeting candidates suppressed	100 / 1
Release to simulated acceptance, 5 s sink delay	100/100 within 180 s, p95 103 s, max 109 s
Release to simulated acceptance, 90 s sink delay	87/100 within 180 s, p95 188 s, max 194 s

The two email-delay runs use the same 100 releases. The 95th-percentile time includes the assumed email delay. Each recipient in this run has one qualifying slot. In UC-02, a student can match several rooms in one cycle. An opening found in the first room could wait another 96 seconds for the last room's read, then need a fresh confirmation. The next experiment should time that whole path through acceptance at a test mailbox.

Decision for P3

Keep one reader with a shared request budget. Run against fixtures until the library answers Q1/Q2 with an approved source and rate. Then check whether one read covers a room's full booking horizon, whether it returns a stable room identifier, slot boundaries, status and capacity, and how the service signals throttling. If the feed requires a separate request for each room and each of seven dates, a sweep alone needs 119 reads against a two-minute allowance of 40. In that case the team needs a bulk feed or a change request before enabling live polling.

The slow-sink run missed QR-01. A pilot should timestamp release and acceptance at a test mailbox and count per-recipient volume before choosing a provider. As of 3 November 2026, Resend listed a free limit of 100 emails a day, exactly the 100 distinct recipients in this fixture. Firebase listed Cloud Messaging at no cost. Account eligibility, delivery to UTD addresses, handling of push subscriptions under QR-06 and real demand remain open for CON-03. Resend pricing · Firebase pricing

Functional trace

For each requirement, the tables show where the behavior occurs, which component owns it, and how to check it. Component IDs C1–C6 and the shared store S1 appear in §6.

ID	Use case and analysis concept	Design owner	Decision	Evidence needed or available
FR-01	UC-01 · Student, Watch	C1, C2	ADR-01	Save a valid watch and reject a date beyond the seven-day horizon.
FR-02	UC-01, UC-03 · Student, Watch	C1, C2	ADR-04	Exercise signed-out create, list and cancel against the OIT-approved NetID exchange. Q3 is open.
FR-03	UC-02 · Room, ReadCycle	C3	ADR-02	§8 measures per-room gaps on fixtures. Q1 and Q2 govern the live source.
FR-04	UC-01, UC-02 · Watch, Slot, Room	C2, C3, C4	ADR-02	Compare known matches with wrong time, date and capacity. The meaning of “bookable” awaits Q4.
FR-05	UC-02 · Slot, AlertItem	C3, C4, C5	ADR-02	Remove a matching slot between its discovery and confirmation. Verify suppression.
FR-06	UC-02 · Student, Alert, AlertItem, DeliveryAttempt	C4, C5, S1	ADR-03	Check a test inbox for all confirmed matches grouped by owner. §8 uses a simulated sink.
FR-07	UC-02 · Room, Slot, Alert	C5	ADR-03	Inspect the name, both times and the library booking link in the received message.
FR-08	UC-01, UC-02 · Watch, DeliveryAttempt	C1, C2, C5	ADR-03, ADR-04	Compare email-only and opted-in watches after the QR-06 subscription decision.
FR-09	UC-03 · Student, Watch, AlertItem	C2, C4, C5, S1	ADR-03, ADR-04	Cancel before dispatch and check the final status and absence of new alerts.
FR-10	UC-01, UC-03 · Student, Watch	C1, C2, S1	ADR-04	Two signed-in owners see only their own Active, Expired and Cancelled watches.
FR-11	UC-02, UC-03 · Watch	C2, C4, C5	ADR-01, ADR-03	Advance past window end, confirm Expired within five minutes and suppress later sends.
FR-12	UC-02 · Watch, Slot, AlertItem	C4, C5, S1	ADR-03	Hold one slot free across ten cycles. Include restart and ambiguous delivery in later checks.
FR-13	UC-01 · Student, Watch	C2, S1	ADR-01	With four Active watches, submit two creates together. Only one may take the fifth place.
FR-14	UC-04 · ReadCycle	C3, C6, S1	ADR-01, ADR-02	During a ten-minute source outage, last success freezes and the failure count rises. Restore the stub and check recovery.

Quality, constraints and gaps

ID	Path and analysis concept	Design owner	Decision	Evidence needed or available
QR-01	UC-02 · Slot, Alert, DeliveryAttempt	C3, C4, C5	ADR-02, ADR-03	§8 measures simulated acceptance. A real email-service timestamp is still needed for QA-1.
QR-02	UC-01 to UC-04 · Watch, ReadCycle	C1, C3, C6	ADR-01	Complete the 14-day opening-hours trial in QA-3.
QR-03	UC-01 · Student, Watch	C1, C2	ADR-01	Time at least ten first-time UTD students on a phone under QA-4.
QR-04	UC-01, UC-03 · Student, Watch	C1	ADR-01	Scan each screen and complete the keyboard and screen-reader walks in QA-4. Q6 is open.
QR-05	UC-02 · Room, ReadCycle	C3	ADR-02	Count all outbound requests on the same clock. §8 exercises the proposed schedule.
QR-06	UC-01 to UC-03 · Student, Watch, DeliveryAttempt	C1, C2, C5, S1	ADR-03, ADR-04	Inspect exports for three test accounts, including stored metadata. Resolve push endpoint retention.
CON-01	UC-01, UC-03 · Student	C1, C2	ADR-04	Obtain OIT direction for NetID-only sign-in and authorized claims (Q3).
CON-02	UC-02 · Room, Slot, ReadCycle	C3	ADR-02	Inspect allowed source requests. The student alone follows the separate library booking link.
CON-03	All use cases · ReadCycle, DeliveryAttempt	C1-C6, S1	ADR-01, ADR-03	Demonstrate that hosting, storage, email and push fit approved zero-cost quotas at the intended load.

9.3 Gaps and their consequences

1. Live availability. Fixture results are conditional on the amount of data one read returns and on available capacity metadata. Without an approved credential-free source and rate, FR-03, FR-04, QR-01 and QR-05 remain unverified against McDermott. Record the Q1/Q2 response and source shape before selecting a live schedule. Q4 separately determines whether a free slot is reservable by a particular student.

2. Delivery and identity. The spike sends after each room confirmation. A student matching more than one room may wait for the full cycle's email batch. At six seconds per room, the last of 17 reads starts 96 seconds after the first. Measure that wait, fresh confirmations and provider acceptance together before judging QR-01. An unknown provider outcome leaves FR-06 in tension with FR-12 (ADR-03). Obtain an OIT decision for FR-02 and CON-01. Keep FR-08 gated until QR-06 is resolved.

3. Human and operating evidence. The repository needs observed usability and accessibility outcomes, a sustained availability trial and measured operating quotas for QR-02 through QR-04 and CON-03. FR-14 remains the sole functional requirement of UC-04 and the first scope reduction proposed by P1 if time is short. Record any cut through change control.

Milestones and working agreement

All dates use Central time. P2 is due 8 November. The P3 work below is scheduled by week until the course deadline is confirmed.

Target	Owner from P1	Output and completion condition
Wed 4 Nov, 7:00–7:30 p.m. CT	Alexander, all owners	First proposed weekly meeting. Review the branch and open questions, assign a backup per item, and agree on the review and meeting cadence.
4–6 Nov	Lee, Mayand	Ask the library and OIT Q1–Q4/Q6 through the team's normal channels. Pilot the survey with enrolled students. Record how they were recruited, when they answered and how many took part.
6–8 Nov	Henry, Arshan, Isaam	Independently walk both sequences, rerun the spike and review the assembled report. One teammate records a substantive PR review before merge. Alexander records the baseline decision.
First P3 week following P2	Henry, Arshan	Detail the ownership and matching paths against a stub. Exercise the full-cycle batch and any fresh confirmation. Identify which failures require an approved requirement change.
Second P3 week	Mayand, Isaam	Compare measured provider acceptance, keyboard access, first-time task completion and free-tier quotas against §3. Start the 14-day availability observation as soon as a trial build is available.
P3 integration, before the confirmed course deadline	All owners, Isaam integrating	Each member commits their own accepted work. Resolve red checks or document approved deviations with issue, reviewer and commit links.

Working agreement. The proposed meeting time is Wednesdays, 7:00–7:30 p.m. CT. Use GitHub Issues for decisions and meeting notes, and pull requests for reviews. The owner proposes a decision. If the team disagrees, the six members vote. A tie or a disagreement that lasts 48 hours goes to the instructor or TA.

Risks and repository practice

Risk	Trigger	Response and owner
R1 · source unavailable or changed	An approved anonymous read cannot return stable room, slot and capacity data.	Arshan keeps the fixture path and asks for a supported bulk feed. Alexander raises a change request if live FR-03/04 cannot be demonstrated.
R2 · rate or permission objection	The library objects, throttles requests or specifies a lower limit.	Lee records the terms. Arshan pauses live reads and recalculates the full read and confirmation budget before any restart.
R3 · NetID access	OIT cannot provide the required sign-in or permitted claims.	Lee records the decision. Alexander proposes a CON-01 change for review. Continue fixture-based identity checks in the meantime.
R4 · incomplete individual evidence	A milestone has no named owner, review, or authored work by its checkpoint.	Isaam asks the owner and backup to split the remaining work, records responsibility, and raises any proposed FR-14 or FR-08 deferral through change control.
R5 · late or unseen alerts	Fewer than 95 of 100 emails reach the service in 180 s, or students report the delay is unhelpful.	Arshan separates source, cycle-end batching and delivery delays. Mayand compares the result with survey evidence before proposing a revised threshold.

Repository. The specifications are in `docs/`, the polling experiment in `docs/spikes/availability/`, and the survey in `docs/elicitation/`. Link each commit to its issue and have a teammate review the pull request before merging.

AI use. OpenAI Codex helped draft and edit the report, draw the diagrams, write the replay and assemble the PDF. Checks included comparing the IDs with P1, rerunning the replay and inspecting the rendered pages.

Open questions and v1.0 status

Question	Evidence and interim decision	Who can close it
Q1 · Supported availability source?	Public inventory exists. A text reader did not obtain timestamped slots for a supported feed. Use fixtures until a response establishes the endpoint, fields and granularity.	Library systems staff and Arshan
Q2 · Automated reads and alerts permitted?	Twenty requests per rolling minute is an internal ceiling. The library has not approved it. Live scheduled polling awaits written terms.	Library reservations staff and Lee
Q3 · NetID and allowed claims?	Design assumes a verified campus subject and email. Exact exchange, enrollment assurance and claim release require OIT confirmation.	OIT and Lee
Q4 · What does bookable mean for one student?	Match the library's free slot and listed capacity. The library decides personal eligibility when the student follows the link.	Library reservations staff and Henry
Q5 · Do the thresholds fit students' experience?	The survey is ready for its pilot. It asks how long students can wait for an alert, how many watches they need and whether a watch should keep running after a match.	Students and Mayand
Q6 · Applicable accessibility standard?	Carry WCAG 2.2 AA forward. The keyboard and screen-reader walks still require a review environment and participants.	OIT/accessibility contact and Mayand

Two questions cut across the plan. Q3 includes whether push subscriptions can be stored under QR-06. R5 includes the time a match spends waiting for the other rooms in its read cycle.

12 Baseline declaration

v1.0 candidate · 3 November 2026. The report, specifications, survey and replay in PR #19 retain FR-01–FR-14, QR-01–QR-06 and CON-01–CON-03.

Each owner checks their section. After an independent teammate reviews the PR, the change-control lead records the approved merge commit and tags it v1.0.

Student survey instrument

Draft 0.1 · 3 November 2026 · Pilot lead: Mayand Dhami

Purpose and collection

Find out how often students have trouble booking a room and how quickly an email would need to arrive for them to act. Recruit at least ten currently enrolled UTD students and record how and when they were recruited. Invite students who use and who do not use the McDermott room calendar. Report those groups separately when describing past booking problems. Use anonymous responses and omit names, email addresses, NetIDs and identifiable free text from the repository. Participation is voluntary and any question may be skipped.

Introduction to read to participants. We are studying how UTD students find group study space at McDermott Library. This survey takes about three minutes. Please describe your own experience. The proposed tool would email you when a matching slot appears. You would still book it on the library's site.

Questions

#	Prompt	Response
S1	Are you currently enrolled at UTD? In the past 30 days, how many times did you try to reserve a McDermott study room?	Enrolled: yes / no / skip. Attempts: 0 / 1 / 2–4 / 5 or more / unsure. End here for respondents who are not enrolled.
S2	Think about your most recent attempt. Was a room available for the time and group size you needed when you first checked?	Yes / no / unsure / no attempt in the past 30 days.
S3	If none was available, what did you do next?	Refreshed the page / checked again later / chose another time / used another space / other / not applicable. Select all that apply.
S4	For an opening that fits your plans, what delay from the opening to an email would still give you time to act?	Under 1 minute / 1–3 minutes / over 3 through 5 / over 5 / would not use an alert / unsure.
S5	Imagine you set a watch for Thursday, 2–4 p.m. It alerts you at 1 p.m. What should the watch do afterward?	Keep watching until the end or until I cancel / stop after the first alert / ask me when I set it / no preference.
S6	During a typical busy week, how many separate room searches would you want active at once?	0 / 1 / 2–3 / 4–5 / 6 or more / unsure.
S7	Which alert would you choose if both were available at no cost?	Email only / email plus browser push / neither / unsure.
S8	What would make this alert unhelpful or difficult to use?	Optional short answer. Please omit identifying information.

How to report the result

1. Record dates, recruitment location or channel, number approached if known, number responding, and eligible responses. Report unanswered items separately. Keep the denominator with each count.

2. Of the students who tried to book a room, count how many found no suitable room on their first check. Of those, count how many checked again. Report counts for each answer to S4–S7. Aggregate S8 themes without publishing raw personal text.

3. Compare the distributions with QR-01's 180-second target, FR-13's five-watch cap and the rule that an alerted watch remains Active. A respondent's answer about tolerable delay measures preference. The separate timed QR-03 study measures whether a first-time user can actually create a watch.

4. The person who runs the survey should commit a dated aggregate report in this directory, link the issue and request a teammate's review. If fewer than ten eligible students respond, report the count and keep the target open.

P1 questions and P3 plan · Quality acceptance criteria · QR-01 issue

Functional acceptance checks

Use the tracked P1 requirements as the acceptance baseline. This guide describes how to verify them and identifies the subset exercised by the P2 spike. Check live-dependent criteria after the matching source, identity or provider is approved. The P2 trace maps each check to its use case and design decision.

Functional checks

Use a controllable library stub, a clock that can be advanced, two distinct student identities, an operator identity, and separate email and push sinks. Capture release, read, confirmation, dispatch and acceptance timestamps on the same clock. Keep test personal information synthetic.

ID	Exercise	Pass condition
FR-01	Sign in and create a watch within the next seven days with a date, time window and group size. Try again with a date eight days ahead.	The list reflects each valid value. The out-of-horizon date is refused with a message.
FR-02	Attempt create, list and cancel separately before and after signing in.	All three signed-out actions send the student to NetID. Each permitted action becomes available after a valid sign-in.
FR-03	Keep the stub reachable for 60 minutes during simulated opening hours. Record each room's first, intermediate and last read.	Every in-scope room is covered, with no gap above 120 seconds, including coverage at the run boundaries.
FR-04	Release a free slot on the watch date, wholly inside its window with enough capacity. Add wrong-date, partially overlapping and too-small controls.	Within one cycle the fitting slot matches and the controls do not.
FR-05	Remove a candidate after discovery and before confirmation.	No alert for the removed slot reaches either sink. A different slot still free at confirmation remains eligible.
FR-06	Release several matches across rooms during one cycle for one owner, plus a match for another owner.	The test inbox for each owner receives exactly one email covering their confirmed matches for that cycle. No other owner's data appears.
FR-07	Inspect the delivered message and follow its link.	Each match lists room name, start and end times. The link opens the library booking page.
FR-08	Enable push for one watch and leave another email-only. Release a distinct matching slot for each.	Both matches appear by email. Only the opted-in match appears in browser push.
FR-09	Cancel an Active watch. After cancellation, release a slot that would match it.	The list shows Cancelled and no new alert for that watch reaches a sink.
FR-10	Under each of two accounts, create, expire and cancel watches. Try to read or change the other owner's watch.	Each account sees only its own watches with correct status. Cross-owner requests reveal no watch details and change nothing.
FR-11	Advance the clock past an Active watch's end. Release a matching slot after the end.	The watch becomes Expired within five minutes and no subsequent match produces an alert.
FR-12	Keep a slot free across ten read cycles for a single watch. Repeat across a worker restart.	Exactly one logical alert covers the pair. A channel accepted before a retry is not resent.
FR-13	With five Active watches, attempt a sixth. Cancel one, then submit another. Also race two creates for the fifth place.	The sixth is refused with a message. A place becomes available after cancellation. At most five are Active under contention.
FR-14	Disable the stub for ten minutes, restore it, then advance the clock through the rolling one-hour failure window.	Last success freezes during the outage, failed reads rise above zero, and the figures recover as new successful cycles complete and old failures age out.

Quality and constraint checks

ID	Evidence and acceptance threshold
QR-01	Under up to 200 Active watches, compare 100 scripted opening times to acceptance timestamps at the student's email service. At least 95 must take 180 seconds or less. Include cycle-end batching, reconfirmation and actual service acceptance.
QR-02	Probe every five minutes during published library opening hours for 14 consecutive days. Count missed probes as failures. At least 99% of scheduled opening-hours checks must succeed.
QR-03	Time at least ten first-time UTD students on phones from the first screen to the new watch appearing in their list. At least 90% must finish in under 90 seconds without help.
QR-04	Evaluate every screen in UC-01 and UC-03 against WCAG 2.2 AA. Require zero serious or critical automated findings and a complete keyboard-only and screen-reader walkthrough. Record manual findings as well as scan results.
QR-05	With 200 Active watches for 24 hours, count every library request, including retries and confirmations. The maximum over any rolling 60 seconds must be at most 20.
QR-06	Review complete data exports for three test students, including linked logs and delivery metadata. The only stored personal data allowed are NetID, email and watches. Resolve push endpoint retention before enabling that channel.
CON-01	Inspect the configured identity provider and exercise all three protected actions. NetID is the sole student sign-in route. Record OIT approval and the attributes actually released.
CON-02	Inspect adapter capabilities and all outbound requests on the stub, including cancellation. No operation creates, changes or cancels a reservation. The student's browser opens the external booking page.
CON-03	Record proposed hosting, storage and delivery quotas, the account configuration, usage at pilot load and any invoice. The team must incur zero cost.

P2 evidence and issue index

The controlled replay exercises the per-room read gap, rolling-minute traffic cap, match filtering, stale-slot suppression, deduplication during a single run and simulated delivery delay. A real provider, a full-cycle batch, restart recovery, NetID and human walkthroughs still require separate evidence.

The issue tracker contained these titled requirements when checked on 3 November. QR-01 appears twice (#1 and #2), as does FR-05 (#16 and #17). Related records are QR-02 #3, QR-03 #4, QR-04 #6, QR-05 #7, QR-06 #8, CON-01–03 #9–11, FR-01–04 #12–15, FR-06 #18 and FR-10 #5. At that check FR-07–09 and FR-11–14 had no matching issue title. The requirements and checks above retain their IDs independently of issue numbering.